Literature Review: Pedagogical Defeasible Reasoning

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Abstract

Defeasible reasoning is a form of non-monotonic reasoning. It allows conclusions to be revised in light of new information. This process is seen as mimicking common sense, some applications include AI reasoning and answer set programming. A more widely known type of logic is classical propositional logic, where the premises assure the conclusion. Given the specialisation of defeasible reasoning there is no pedagogical defeasible reasoning tools to assist in learning/understanding it compared to the more known classical reasoning. In this literature review we aim to discuss the two logics, building up the tools to define them and explore some of their algorithms. Additionally we observe and analysis some existing pedagogical tools for logic to aid our development of a pedagogical tool for defeasible reasoning for rational, lexicographical and relevant closure.

1 Introduction

Defeasible reasoning is a type of reasoning that revises the conclusion in light of new information [14]. The conclusion is a consequence of some statements in a knowledge base where external new information causes it to be reconsidered. It is a more realistic approach to modelling the world compared to other forms of logic. This approach to reasoning is termed "common sense", it closely mimics human thinking, such as deciding the applicability of scientific models [18] . To the best of our knowledge there is no pedagogical defeasible reasoning tools available. Given its importance in the growing field of artificial intelligence it would prove beneficial to produce such a pedagogical tool. There are different ways humans think and apply information and this is true in the different defeasible entailment, there is a range consisting of rational, lexicographical and relevant. To understand these algorithms we explore classical propositional logic first to define the foundation which defeasible reasoning is built on.

Classical propositional logic is the foundation on which all other logics are built on [2]. This would be our starting point, developing the language and syntax needed to understand defeasible reasoning. Propositional logic is the formalization of reasoning about knowledge [13] . Through its language and syntax we create statements with truth values. We use the statements in conjunctions with boolean operators to form more complicated statements, formulas [2]. This allows us to reason about the statements and determine what can be derived from said statements [13]. Unlike with defeasible reasoning the conclusion has to be a consequence of the statements inference. This property is known has monotonicity.

There are several approaches we can take to defining defeasible reasoning. In this literature review we will consider KLM preferential reasoning[15] framework that uses propositional logic as a base. The framework defines the postulates required for entailment and introduces a preferential approach that ranks possible worlds. In this literature review we will consider rational closure[17], lexicographical closure[16] and relevant closure[4]. The focus would be on relevant closure, using the other two closures to contrast differences. We will build up each closure from a algorithmic perspective, showing the approach they take to defeasible entailment.

Pedagogical tools are alternatives to traditional learning where participation induces active learning which has shown to better suit students. Logic has been around for a long time and still serves a vital purpose in understanding and programming systems. To our knowledge no pedagogical defeasible reasoning tool exists. In this literature review we will analyse previous work done pertaining to logical tools, to aid our development of a defeasible reasoning tool.

The layout of the paper is as follows, Section 2 will introduce propositional logic, where we define the syntactic and semantic elements working our way to classical entailment. In section 3 we begin our discussion on defeasible entailment, using the preferential approach in this literature review. We will work our way from an outline of preferential semantics to rational closure[17], lexicographical closure[16] and finally relevant closure[4]. Section 4 will assess the previous work done relating to logical tools, we will assess their functions and limitations with regard to our prospective defeasible reasoning tool.

2 Propositional Logic

Propositional logic provides us a simple system consisting of atomic propositions, boolean operators and formulas. These components provide us a formal method to reason[2]. Understanding the syntax and semantics used in propositional logic provides the foundation needed for KLM framework[15] and the algorithms used to explain defeasible entailment.

2.1 Syntax

Atomic propositions, shorthanded to atoms are the furthest we can decompose a statement, an example would be birds or fly. We can denote atoms with lowercase letters like so b : Birds. Where b now represents the statement. Boolean operators are operators that join atoms together. These operators include:

- negation $\neg$
- conjunction $\wedge$
- disjunction $\vee$
- implication $\rightarrow$
- equivalence $\leftrightarrow$

The operators we consider consists of a unary negation operator while the rest are binary. Formulas are defined as consisting of one or more atoms connected by boolean operators. An atom on its own can be a formula[2], some examples of formulas include:

- b
- $b \rightarrow f$

- $\neg f$

2.2 Object Level Semantics

To give meaning to our logic we assign atoms truth values T for true and F for false. We recursively define a formula's truth value as a result of its constituents. The function that assigns the truth values is called an valuation, defined as $u : \mathcal{P} \rightarrow \{T, F\}$ [13]. We can use a truth table to conveniently show our formulas with their respective possible interpretations as follows:

Table 1: Truth Table for Boolean Operators

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

2.3 Meta semantics

If the valuation makes a atom true we say that it satisfies the atom. We denote this as $u \Vdash q$ where for some valuation u, satisfies q, $u(q) = T$. A valuation u that results in a formula p being satisfied is referred to as a model.

We can extend satisfaction to formulas, a formula is satisfied by the valuation u if all its constituents are satisfied. Likewise a knowledge base $\mathcal{K}$ is satisfied iff every $a \in K$ it is the case $u \Vdash a$.

Classical entailment can be defined over the language, it is a meta level operator as it serves to infer about the statements in the knowledge base. $q \models p$ which reads p is a logical consequence of q iff $u \Vdash q$ then $u \Vdash p$. We extend this to a knowledge base $\mathcal{K}$ and α a formula, we say $\mathcal{K}$ entails α denoted, $\mathcal{K} \models \alpha$ iff every model of $\mathcal{K}$ is also a model of α. In natural language this means there is no world where the statements in $\mathcal{K}$ are all true and α is false.

Monotonicity is a property of classical reasoning, it asserts that inference can only be added to knowledge base not retracted. This property weakens classical entailment as it is unable to differentiate exceptions from the general statements. Consider knowledge base $\mathcal{K} = \{b \rightarrow f, p \rightarrow b, p \rightarrow \neg f\}$, we have that both "$p \rightarrow \neg f$" and "$p \rightarrow f$" via transitivity. This shows that p can not exist, which is undesirable.

3 Defeasible Reasoning

We now turn our attention to non-monotonic reasoning as a solution.

3.1 Preferential Approach

The preferential approach to non-monotonic reasoning was first defined by [21, 22]. This approach introduced preference for valuations, a formula is thus satisfiable if the the most preferred models are satisfied [13]. The set of most preferred models is defined as the models that are minimal with respect to the preference relation[13]. [8] developed a proof system for preferential reasoning which was extended by KLM[15].

[15] and [17] had defined an extension to propositional logic. The extension enabled us to capture a state where knowledge is assumed typical until exceptional information says otherwise. In this literature review we will consider preferential reasoning as it has been described as the core of non-monotonic reasoning [5].

3.2 KLM

The KLM framework[15] denotes $|\sim$ such that $a \ |\sim \ b$ reads b is a preferential consequence of a unless there is exceptional information stating otherwise. These preferential semantics add preference interpretations, we say the valuations are classified by typicality [22]. The most typical valuations are more preferred. This ordering is used to develop ranked interpretations where instead of using a graph we use a number of tiers to visualize the interpretations.

A ranked interpretation is defined by using a function $\mathcal{R} : \mathcal{U} \rightarrow \mathcal{N} \cup \{\infty\}$ to define the ordering of a valuation u, where the lower the number the more typical the valuation and the higher the number the more improbable.

Preferential entailment changes $|\sim$ to move from meta level semantics to object level semantics, to explicitly specify defeasible information. Preferential entailment works analogously to classical entailment defined as $\mathcal{K} \ |\approx \ a \ |\sim \ b$ where $|\sim$ becomes the implication for defeasible information, reading "The knowledge base defeasibly entails a typically implies b".

We can combine the preferential entailment and ranked interpretation to define ranked entailment. Because we consider all the rank interpretations in ranked entailment it is monotonic since information is not retracted so we look to minimal ranked entailment as a solution for non-monotonic reasoning.

We define minimal ranked entailment as $|\approx$ such that for any defeasible implication $a \ |\sim \ b$, $\mathcal{K} \ |\approx \ a \ |\sim \ b$ if and only if $\mathcal{R}^K_{RC} \Vdash a \ |\sim \ b$, where $\mathcal{R}^K_{RC}$ is the minimal ranked interpretation in the set. To show minimal ranked entailment is non-monotonic consider $\mathcal{K} = \{p \ |\sim \ b, b \ |\sim \ f, p \ |\sim \ \neg f\}$, The minimal ranked model of the knowledge base: We first consider exceptionality in the knowledge to rank. We find that $b \ |\sim \ f$ is exceptional since it conflicts with the others rules since $p \ |\sim \ b, p \ |\sim \ \neg f$, so we push its rank up, resulting in:

1	$b \ |\sim \ f$
0	$p \ |\sim \ \neg f, p \ |\sim \ b$

Table 2: Minimal ranked interpretation $\mathcal{R}^{RC}_{\mathcal{K}}$

Now we are able to ask "Do penguins fly". The rank of a penguin that flies is 1 compared to the rank of a penguin that does not fly being 0. We conclude that the lower rank is most typical so no penguins do not fly.

3.3 Rational Closure

Rational closure is an alternative syntactic definition of minimal ranked entailment [13]. It is a non-monotonic closure that satisfies LM-rationality [17]. Being non-monotonic means allowing exceptional information to change the conclusion.

Materialisation is the propositional formula, known as the material counterpart of a defeasible implication. An example would be:

Defeasible implciation	Material counterpart
$a \mid\sim b$	$a \rightarrow b$

Table 3: Materialisation

Likewise the same is done for a defeasible knowledge base $\mathcal{K}$ which has material counterpart $\overrightarrow{\mathcal{K}}$. Materialisation is used to check exceptionality in rational closure. We show how the material counterpart relates to the defeasible part.

$$\overrightarrow{\mathcal{K}} \models a \text{ iff } \mathcal{K} \mid\approx \top \mid\sim a \tag{1}$$

We then define exceptionality[17] for a propositional formula a over a non-defeasible knowledge base $\mathcal{K}$ with meta level semantics:

$$a \text{ is exceptional for } \mathcal{K} \text{ iff } \mathcal{K} \mid\approx_P \top \mid\sim_P \neg a \tag{2}$$

Combining the two given defeasible knowledge base $\mathcal{K}$:

$$a \text{ is exceptional for } \mathcal{K} \text{ iff } \overrightarrow{\mathcal{K}} \models \neg a \tag{3}$$

This allows us to reduce the exceptionality check to a classical entailment check over $\overrightarrow{\mathcal{K}}$, making the rational closure algorithm computable using classical reasoning[17]. Consider example:

$$\mathcal{K} = \{b \mid\sim f, p \mid\sim b, p \mid\sim \neg f\}$$

$$\overrightarrow{\mathcal{K}} = \{b \rightarrow f, p \rightarrow b, p \rightarrow \neg f\}$$

We can check if b is exceptional using $\overrightarrow{\mathcal{K}} \models \neg b$ since $b \rightarrow f$ leads to the contradiction $p \rightarrow f$ we conclude that $\overrightarrow{\mathcal{K}} \models \neg b$ follows showing that b is exceptional.

Lehmann and Magidor [17] showed how we define exceptional information, a formula is exceptional if its negation is typical in every preferential interpretation. Exceptionality allows us to rank the knowledge base. We define a function ε to return a subset of the knowledge base where every defeasible statement with an exceptional antecedent is returned, Formally:

$$\varepsilon(\mathcal{K})\{\alpha \mid\sim \beta \mid \mathcal{K} \mid\approx_P \top \mid\sim \neg\alpha\}$$

[13].

Using the above mentioned function we can create a sequence of knowledge bases: $\mathcal{E}_i^K$ where K is the knowledge base and i is the rank.

$$\mathcal{E}_0^{\mathcal{K}}\mathcal{K} \tag{4}$$

$$\mathcal{E}_i^{\mathcal{K}}\varepsilon(\mathcal{E}_{i-1}^{\mathcal{K}}) \tag{5}$$

$$\mathcal{E}_\infty^{\mathcal{K}}\emptyset \tag{6}$$

This sequence is constructed by repeatedly applying the exceptionality function until a fixpoint is reached, that is until $\mathcal{E}_i^{\mathcal{K}} = \mathcal{E}_{i-1}^{\mathcal{K}}$. Since $\mathcal{K}$ is finite this sequence is guaranteed to terminate [17]. We note that $\mathcal{E}_\infty^{\mathcal{K}}\emptyset$ if there is no formula that is exceptional at every level, if $\mathcal{E}_\infty^{\mathcal{K}}$ exists than it represents formulas that are impossible and iff the classical formula or defeasible form $\neg a \mid\sim$ has $br(\neg a) = \infty$ it will be in the knowledge base. Using the example for materilisation we would get:

$$\mathcal{E}_0^{\mathcal{K}} = \{p \mid\sim b, p \mid\sim \neg f, b \mid\sim f\}$$

$$\mathcal{E}_1^{\mathcal{K}} = \{b \mid\sim f\}$$

We define base rank to be $br(\alpha)$ is the smallest r such that α is not exceptional in $\mathcal{E}_r^K$. The base rank shows at what level α stops being exceptional. Formally:

$$br_{\mathcal{K}}(\alpha) \min\{r \mid \mathcal{E}_r^{\mathcal{K}} \mid\not\approx \top \mid\sim \neg\alpha\} \tag{7}$$

Additionally the base rank for a defeasible implication is the base rank for the antecedent:

$$br_{\mathcal{K}}(\alpha \mid\sim \beta) \equiv br_{\mathcal{K}}(\alpha) \tag{8}$$

The idea behind base rank is that the lower the rank the more general a statement is thus more defeasible and likely for exceptional specific knowledge to conflict[13]. Using the example for the sequence of knowledge bases we can calculate the base ranks as follows:

$$br_{\mathcal{K}}(b \mid\sim f) = br_{\mathcal{K}}(b) = 1$$

$$br_{\mathcal{K}}(p \mid\sim \neg f) = br_{\mathcal{K}}(p) = 0$$

$$br_{\mathcal{K}}(p \mid\sim b) = br_{\mathcal{K}}(p) = 0$$

Using the definition of base rank we can construct a non-monotonic entailment called rational closure[17]. The semantic definition is that of minimal ranked entailment [10] .

$$\mathcal{K} \mid\approx_{RC} \alpha \mid\sim \beta \text{ iff } br_{\mathcal{K}}(\alpha) < br_{\mathcal{K}}(\alpha \wedge \neg\beta) \text{ or } br_{\mathcal{K}}(\alpha) = \infty \tag{9}$$

Once again considering the example used in materialisation that we have appleid over base ranks we can check if $\mathcal{K}$ entails "$p \mid\sim \neg f$":

$$br_{\mathcal{K}}(p) < br_{\mathcal{K}}(p \wedge \neg(\neg f))$$

$$0 < br_{\mathcal{K}}(p \wedge f)$$

$$0 < \overrightarrow{\mathcal{E}_0^{\mathcal{K}}} \models \neg(p \wedge f)$$

$$0 < \overrightarrow{\mathcal{E}_0^{\mathcal{K}}} \models p \rightarrow \neg f \text{ exceptional}$$

$$0 < \overrightarrow{\mathcal{E}_1^{\mathcal{K}}} \models p \rightarrow \neg f \text{ exceptional}$$

$$0 < \infty$$

Thus

$$\mathcal{K} \mid\approx_{RC} p \mid\sim \neg f$$

this particular approach can be implemented purely with a classical entailment SAT solver [17].

3.4 Rational closure Algorithm

Now we will consider the following algorithms, BaseRank and Rational closure.

The BaseRank algorithm is responsible for generating a output tuple representing the material counterparts of defeasible implications corresponding to the sequence of exceptioonal subsets to $\mathcal{K}$ [13]. The algorithm takes a defeasible knowledge base as input, converting it into a material counterpart , $\overrightarrow{\mathcal{K}}$, on which most computation will occur. We begin by building the exceptional subsets of $\overrightarrow{\mathcal{K}}$, we then find the difference between the current and next exceptional subset, $\mathcal{E}_i \setminus \mathcal{E}_{i+1}$, to store them as a unique ranked instance, R_i in the tuple. After all propositional formulas have been assigned into a ranked instance we calculate the sum of ranked instances to include alongside the ranked instances in the tuple.

The RationalClosure algorithm returns a true or false answer with regard to a defeasible knowledge base entailing query. The algorithm incorporates the BaseRank algorithm therefore the inputs are the same. It then creates a union of all the output sets in order

Algorithm 1 BaseRank

Input: A defeasible knowledge base $\mathcal{K}$
Output: An ordered tuple $(R_0, \ldots, R_{n-1}, R_\infty, n)$
$i \leftarrow 0$
$E_0 \leftarrow \overrightarrow{\mathcal{K}}$
while $E_{i-1} \neq E_i$ **do**
    $E_{i+1} \leftarrow \{\alpha \rightarrow \beta \in E_i \mid E_i \models \neg\alpha\}$
    $R_i \leftarrow E_i \setminus E_{i+1}$
    $i \leftarrow i + 1$
end while
$R_\infty \leftarrow E_{i-1}$
if $E_{i-1} = \emptyset$ **then**
    $n \leftarrow i - 1$
else
    $n \leftarrow i$
end ifreturn $(R_0, \ldots, R_{n-1}, R_\infty, n)$

and using classical entailment iterates through the ranked instances until the query is not exceptional. Once a rank is found it returns if the query is classically entailed. [7] has shown that a query $a \mid\sim b$ and defeasible knowledge base $\mathcal{K}$ returns true iff $a \mid\sim b$ is in the rational closure of $\mathcal{K}$ [13].

Algorithm 2 RationalClosure

Input: A defeasible knowledge base $\mathcal{K}$, and a defeasible implication $\alpha \mid\sim \beta$
Output: *true*, if $\mathcal{K} \mid\approx_{RC} \alpha \mid\sim \beta$, and *false* otherwise
$(R_0, \ldots, R_{n-1}, R_\infty, n) \leftarrow$ BaseRank($\mathcal{K}$)
$i \leftarrow 0$
$R \leftarrow \bigcup_{j=0}^{n} R_j$
while $R_\infty \cup R \models \neg\alpha$ **and** $R \neq \emptyset$ **do**
    $R \leftarrow R \setminus R_i$
    $i \leftarrow i + 1$
end whilereturn $R_\infty \cup R \models \alpha \rightarrow \beta$

To consolidate the algorithms consider example:

$$p : \text{penguin}$$
$$f : \text{fly}$$
$$b : \text{bird}$$
$$w : \text{wings}$$
$$\mathcal{K} = \{p \mid\sim b, b \mid\sim f, p \mid\sim \neg f, b \mid\sim w\}$$
$$\overrightarrow{\mathcal{K}} = \{p \rightarrow b, b \rightarrow f, p \rightarrow \neg f, b \rightarrow w\}$$

Applying the BaseRank algorithm

$$\mathcal{E}_0 = \overrightarrow{\mathcal{K}}$$
$$\mathcal{E}_1 = \{p \rightarrow \neg f, p \rightarrow b\}, (\mathcal{E}_0 \models \neg p)$$
$$R_0 = \{b \rightarrow f, b \rightarrow w\}$$
$$R_1 = \{p \rightarrow b, p \rightarrow \neg f\}$$
$$R_\infty = \emptyset$$
$$n = 3$$

Producing tuple:

$$(\{b \rightarrow f, b \rightarrow w\}, \{p \rightarrow b, p \rightarrow \neg f\}, \emptyset, 3)$$

R_∞	$\emptyset$
R_1	$p \rightarrow b, p \rightarrow \neg f$
R_0	$b \rightarrow f, b \rightarrow w$

Table 4: BaseRank

Now in the rational closure algorithm, consider: Because $R_0 \cup R_1 \cup R_\infty \models \neg p$ we remove R_0 and are left with:

R_∞	$\emptyset$
R_1	$p \rightarrow b, p \rightarrow \neg f$

Table 5: BaseRank

$R = R_1$ Queries: "$p \mid\sim f$":

$R_\infty \cup R \not\models p \rightarrow f$, (return false, was contradicted by $p \rightarrow \neg f$)

"$p \mid\sim \neg f$":

$$R_\infty \cup R \models p \rightarrow \neg f, \text{(return true)}$$

"$p \mid\sim w$" :

$$R_\infty \cup R \not\models p \rightarrow w, \text{(return false } b \rightarrow w \text{ removed)}$$

The last query illustrates the drowning problem. Where properties meant to be inherited are not. Considering a query related to b now, we see that $R_0 \cup R_1 \cup R_\infty \not\models \neg b$ so $R = R_0 \cup R_1$ and the resulting table is the same as in Table4. we now consider query $b \mid\sim f$:

$$R_\infty \cup R \models b \rightarrow f, \text{(return true)}$$

3.5 Lexicographical Closure

Lexicographical Closure is another defeasible entailment relation proposed by [16]. It is a more liberal entailment than rational closure, allowing for the inheritance of generic properties. Lehmann defines the properties: presumption of typicality; presumption of independence and priority of typicality to guide the definition of lexicographical closure. It has been shown that lexicographical closure is a rational entailment [5] as we will see in the algorithm.

3.6 Lexicographical Closure Algorithm

We will now consider the algorithms used for lexicographical closure, adapted for propositional logic by [5]. The algorithm, SubsetRankProp uses a new ranking of statements by using the ranks computed by the BaseRank algorithm and adding new ranks $D_{i,j}$ between the existing ranks R_i. $D_{i,j}$ represents all the ways to represent statements in R_i [20] . Once a new ranking of statements is complete we return the tuple and use the RationalClosure Algorithm to compute the rest of the LexicographicalClosure Algorithm.

We will be examining the combined formula approach to lexicographical closure thus we must understand some important fundamental formulas [12]:

(1) $\mathcal{K} \cup \{\phi \vee \psi\} \models \chi$ if and only if $\mathcal{K} \cup \{\phi\} \models \chi$ and $\mathcal{K} \cup \{\psi\} \models \chi$
(2) $\mathcal{K} \cup \{\phi \wedge \psi\} \models \chi$ if and only if $\mathcal{K} \cup \{\phi,\ \psi\} \models \chi$

Taking the Base rank table produced in Table 4. We can apply SubsetRankProp to the statements as follows:

Algorithm 3 SubsetRankProp

Input: A defeasible propositional knowledge base $\mathcal{K}$
Output: An ordered tuple $(R_0, \ldots, R_k, R_\infty, k+1)$

$(B_0, \ldots, B_{m-1}, B_\infty, m) \leftarrow$ BaseRankProp($\mathcal{K}$)
$i \leftarrow 0$; $k \leftarrow 0$
repeat
  for $j \leftarrow |B_i|$ **downto** 1 **do**
    $S_{i,j} \leftarrow$ Subsets(B_i, j)
    $D_{i,j} \leftarrow \bigvee_{X \in S_{i,j}} \bigwedge_{x \in X} x$
    $R_k \leftarrow D_{i,j}$
    $k \leftarrow k+1$
  end for
  $i \leftarrow i+1$
until $i = m$
$R_\infty \leftarrow B_\infty$
return $(R_0, \ldots, R_k, R_\infty, k+1)$

Algorithm 4 LexicographicalClosure

Input: A defeasible knowledge base $\mathcal{K}$, and a defeasible implication $\alpha \mid\sim \beta$
Output: *true*, if $\mathcal{K} \mid\approx_{RC} \alpha \mid\sim \beta$, and *false* otherwise
$(R_0, \ldots, R_{n-1}, R_\infty, n) \leftarrow$ SubsetRankProp($\mathcal{K}$)
$i \leftarrow 0$
$R \leftarrow \bigcup_{j=0}^{n} R_j$
while $R_\infty \cup R \models \neg\alpha$ **and** $R \neq \emptyset$ **do**
  $R \leftarrow R \setminus R_i$
  $i \leftarrow i+1$
end whilereturn $R_\infty \cup R \models \alpha \rightarrow \beta$

B_∞	$\emptyset$
B_1	$p \rightarrow b, p \rightarrow \neg f$
B_0	$b \rightarrow f, b \rightarrow w$

Table 6: Ranking

Starting with B_0 we construct subsets of size 2 and 1 resulting as $S_{0,2} = \{\{b \rightarrow f, b \rightarrow w\}\}$, $D_{0,2} = (b \rightarrow f) \wedge (b \rightarrow w)$, $R_0 = D_{0,2}$. We then do the same for $S_{0,1} = \{\{b \rightarrow f\}, \{b \rightarrow w\}\}$,$D_{0,1} = (b \rightarrow f) \vee (b \rightarrow w)$, $R_1 = D_{0,1}$. We do the same for the remaining B_1 and B_∞ :

B_∞	$\emptyset$
$R_3 = D_{1,1}$	$(p \rightarrow b) \vee (p \rightarrow \neg f)$
$R_2 = D_{1,2}$	$(p \rightarrow b) \wedge (p \rightarrow \neg f)$
$R_1 = D_{0,1}$	$(b \rightarrow f) \vee (b \rightarrow w)$
$R_0 = D_{0,2}$	$(b \rightarrow f) \wedge (b \rightarrow w)$

Table 7: SubsetRanking

We then use our subset ranking in 7 as the input for rational closure with query "$p \mid\sim w$" to test: We remove R_0 since $R_\infty \cup R \models \neg p$ for $i = 0$. We test i =1, $R_\infty \cup R \not\models \neg p$ so we test if "$p \rightarrow w$" is entailed by $R_\infty \cup R$ and it is , we check $R_\infty \cup R \models p \rightarrow w$ so we return true. We have now shown how lexicographical closure has fixed the drowning problem.

3.7 Relevant Closure

Relevant Closure is another defeasible entailment relation and solution to the drowning problem. It prevents statements that have no conflict with the conclusion from being thrown out. [20] states that removing a entire level of statements is excessive so relevant closure takes a finer-grain approach to remove the statements causing the conflict in that level. Relevant closure is defined by [4] in a description language thus we use [20] translation of the algorithm into propositional logic. There are two algorithms for relevant closure, basic relevant closure and minimal relevant closure in this literature review we consider minimal relevant closure for reasons discussed below.

3.8 Relevant Closure Algorithm

Algorithm 5 RelevantClosureProp

Input: A defeasible propositional knowledge base $\mathcal{K}$, a defeasible implication $\alpha \mid\sim \beta$, and a partition $\langle R, R^- \rangle$ of $\mathcal{K}$
Output: *true*, if $\mathcal{K} \mid\approx \alpha \mid\sim \beta$, and *false* otherwise

$(R_0, \ldots, R_{n-1}, R_\infty, n) \leftarrow$ BaseRankProp($\mathcal{K}$)
$i \leftarrow 0$
$R' \leftarrow R$
while $R_\infty \cup R^- \cup R' \models \neg\alpha$ **and** $R' \neq \emptyset$ **do**
  $R' \leftarrow R' \setminus \{R_i \cap R\}$
  $i \leftarrow i+1$
end while
return $R_\infty \cup R^- \cup R' \models \alpha \rightarrow \beta$

We now need to understand the notion of relevance, to understand the partition $\langle R, R^- \rangle$. We say R is the set of all information relevant to a query and therefore will only consider statements in R removable. [4] states the notion of relevance is based on justification.

Definition 1. *α is said to be exceptional for $\mathcal{K}$ if $\overrightarrow{\mathcal{K}} \models \neg\alpha$.*

Definition 2. *Let $\mathcal{K}$ be a knowledge base, $J \subseteq \mathcal{K}$ such that J only contains defeasible implications, and α a propositional sentence. Then J is said to be an α-justification w.r.t. $\mathcal{K}$ if α is exceptional for J and for any $J' \subset J$, α is not exceptional for J'.*

Definition 3. *For a sentence α and knowledge base $\mathcal{K}$, let $\mathcal{J}_\mathcal{K}(\alpha) = \{J \mid J$ is an α-justification w.r.t. $\mathcal{K}\}$. Then $\alpha \mid\sim \beta$ is said to be in the Basic Relevant Closure of $\mathcal{K}$ if it is in the Relevant Closure of $\mathcal{K}$ w.r.t. $\bigcup \mathcal{J}_\mathcal{K}(\alpha)$.*

[20] argues that using basic relevant closure causes too many statements to be relevant to the query because we are considering all the statements in a justification. As a result we consider the minimal ranked approach where the lowest ranked statements in a justification are selected as relevant only.

Definition 4. *For some set of justifications $J \subseteq \mathcal{K}$, let $J^{min}_\mathcal{K} = \{\alpha \mid\sim \beta \mid r_\mathcal{K}(\alpha) \leq r_\mathcal{K}(\gamma)$ for every $\gamma \mid\sim \lambda \in J\}$. For a sentence α, let $J^{min}_\mathcal{K}(\alpha) = \bigcup_{J \in \mathcal{J}_\mathcal{K}(\alpha)} J^{min}_\mathcal{K}$. Then $\alpha \mid\sim \beta$ is said to be in the*

Minimal Relevant Closure of $\mathcal{K}$ if it is in the Relevant Closure of $\mathcal{K}$ w.r.t. $\bigcup J^{min}_{\mathcal{K}}(\alpha)$.

We define $R = \bigcup J^{min}_{\mathcal{K}}(\alpha)$ to hold the minimum amount of relevant statement to some query and $R^- = K \setminus R$ the irrelevant statements.

Lets consider $\mathcal{K} = \{p \mid\sim b, b \mid\sim f, p \mid\sim \neg f, b \mid\sim w\}$, query "$p \mid\sim w$" We find the base rank:

1	$p \mid\sim b, p \mid\sim \neg f$
0	$b \mid\sim w, b \mid\sim f$

Table 8: Base Rank

Then find the justifications, $J_{\mathcal{K}}(p)$ for K. We start with $\mathcal{J} = \mathcal{K}$, $\overrightarrow{\mathcal{J}} \models \neg p$ remains true so we try removing each formula, we find that we can remove $b \mid\sim w$ and $\overrightarrow{\mathcal{J}} \models \neg p$ will still be true, there are no other formulas we can remove. We now have $\overrightarrow{\mathcal{J}} = \{p \mid\sim b, b \mid\sim f, p \mid\sim \neg f\}$. Next we check $\overrightarrow{\mathcal{J}}_{min}(p)$ we find the formulas of the lowest base rank8. $\overrightarrow{\mathcal{J}}_{min}(p) = \{b \mid\sim f\}$. We now have $R = \overrightarrow{\mathcal{J}}_{min}(p) = \{b \mid\sim f\}$ and $R^- = \mathcal{K} \setminus R = \{b \mid\sim w, p \mid\sim \neg f, p \mid\sim b\}$. Executing the Rational closure algorithm we get a ranking similar to 4.

While executing in the while loop the iteration for $i = 0$, $R_\infty \cup R^- \cup R' \models \neg p$,is true, so we remove $R_0 \cap R = \{b \mid\sim f\}$ from R', for the next iteration $i = 1$ we now have $R_\infty \cup R^- \cup R' \models \neg p$ being false, we end the loop and check $R_\infty \cup R^- \cup R' \models p \mid\sim w$ and it is true.

4 Pedagogical Tools

Pedagogical tools are designed to support and enhance the understanding of a complex or abstract system. [23] talks about the notion of mathematical and logical anxiety as a result of logic's complex notation and abstract semantics. Given how well established classical propositional logic is, a number of pedagogical tools have been developed to aid with understanding it. We consider previous works [3, 6, 24] [11] [19] that have attempted to use frameworks to explain logic. We will analysis these works to aid our development of a pedagogical tool for defeasible reasoning. In this literature review we are concerned with the explanation of logic rather than effectively teaching.

4.1 Tweety

Tweety [24] is a collection of Java libraries that provides a general interface layer for doing research in and working with different knowledge representation. It serves as a common framework for implementing different approaches to logical approaches of artificial intelligence in general and knowledge representation. Tweety implements defeasible logic programming via DeLP [9] which is different from the KLM[15] preferential reasoning we will be working with. There is no rational or relevance closure built into the TweetyProject.

4.2 Logicraft

Logicraft [6] is a game based educational framework for learning propositional logic. Logic has a abstract nature making it harder to visualize, this alienation makes learners less motivated and engaged [23] . The framework uses puzzle-based challenges where students apply propositional logic rules to progress. It implements various pedagogical approaches such as game based learning and immediate feedback. [6] is limited to propositional logic and given the nature of defeasible reasoning would not be expandable into the defeasible reasoning domain.

4.3 Incredible proof machine

[3] is a interactive theorem prover design for high school learners. The learner drags blocks into a graph to complete a proof in propositional logic. Whilst completing the proof the user is informed of how optimal their solution is to disincentivising bloated proof construction. [3] is limited to propositional logic, however its drag and drop framework is extendible to a defeasible approach.

4.4 RaMP

RaMP[19] is a plugin for the Protégé ontology editor. The plugin implements prototypical and presumptive reasoning over ontologies, enabling logical inferencing to determine defeasible conclusions over defeasible information in the ontology. The plugin represents defeasible information using axiom annotations in Protégé, allowing users to indicate which axioms are defeasible and rank them. RaMP is able to visualise the axiom rankings which our tool also aims to do, it implements prototypical and presumptive reasoning which relates to the closures we consider in this literature review. The plugin operates in Protégé using description logic instead of propositional logic aimed to be more of a research tool.

4.5 SILK

SILK [11] is a expressive semantic rule language that uses courteous prioritized defeasible reasoning. It is a more extensive GUI functional system than previous works [1]. SILK implements a graphical user interface, knowledge entry, query answering and justification browsing for defeasible information as a plugin for the Eclipse IDE. The plugin allows what if exploration through its automatic tracking of target query results against user changes to rules. The justification is implemented as a tree enabling deep exploration of the knowledge to understand the query. SILK is fundamentally different from the intended prospective tool we hope to develop, it uses manual ranking and not KLM preferential reasoning. However, there is functionality that we would consider for our defeasible reasoning tool.

4.6 Gap

We have discussed various logical tools centred around logic. The least relevant tool is Tweety [24], well there is defeasible reasoning implementations it is in DeLP and in general Tweety serves more as a research tool rather than pedagogical. [3, 6] have tools both for pedagogical purposes but they are catered for propositional logic, this could make it easier to implement a defeasible extension. [11] has a wider array of functionality, using a GUI the user has access to justification browsing which is a functionality we would

look to implement. it falls short with it's manual ranking. [19] is the most relevant tool referenced. It is a defeasible plugin for the Protégé ontology editor. The tool provides prototypical and presumptive reasoning over ontologies with a visualisation of axiom rankings. There is also a GUI for checking entailments with immediate feedback. This tool offers the most promising features we would reference. It should be said that well there are implementations of defeasible tools they are restricted to different frameworks or made for research purposes.

5 Conclusion

In this literature we started by defining the foundational logic of KLM[15] that is propositional logic. We defined the syntax and semantics building towards classical entailment, where we discussed its shortcomings due to its monotonic property. This lead us to explore the non-monotonic alternative, defeasible entailment.

We outlined KLM preferential reasoning[15] as the framework this literature review will consider. Using preferential semantics we discuss how to rank valuations. We then defined the various defeasible entailment relations rational closure, lexicographical closure and relevant closure. We discussed how rational closure was too conservative leading to the discarding of properties that should of been inherited, also known as the drowning problem. Lexicographical closure provided a solution through an alternate ranking algorithm known as SubsetRankingProp, which removes the specific statements causing a conflict instead of the whole rank. Relevant closure introduced the idea of relevance which demarcated which statements are relevant to a query, this allowed us to keep statements that are irrelevant to the conflict and only remove statements relevant to the conflict of a query.

Having defined the defeasible entailments we looked at the available logical tools to analysis what was currently being offered. We discussed five logical tools that provided insight into how we can develop our defeasible reasoning tool. The tools ranged from defeasible reasoning research tools to pedagogical logic tools catered to propositional logic. We concluded that these tools each have features that would be functionally relevant and would be incorporated.

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